



Unprovoked seizures after traumatic brain injury: A population-based case–control study

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SUMMARY

Objective: To quantify the risk of unprovoked seizures after traumatic brain injury (TBI)

Methods: We used the Stockholm Incidence Registry on Epilepsy to carry out a population-based case–control study, including 1,885 cases with incident unprovoked seizures from September 1, 2000 through August 31, 2008, together with 15,080 matched controls. Information of prior hospitalizations for TBI was obtained through record linkage with the Swedish National Inpatient Registry for the period 1980–2008. Relative risks (RRs) for unprovoked seizures were estimated after various TBI diagnoses, and influences of TBI severity and time since trauma were studied in detail.

Results: After hospitalization for mild TBI, the RR was 2.0 (95% confidence interval [CI] 1.5–2.7). The RR was higher after brain contusion (5.9, 95% CI 2.4–15.0) or intracranial hemorrhage (ICH) (4.5, 95% CI 2.2–9.0), whereas a combination of both diagnoses led to a further sevenfold increase in RR (42.6, 95% CI 12.2–148.5). The risk was greatest during the first 6 months after severe TBI (RR 48.9, 95% CI 10.9–218.9) or mild TBI (RR 8.1, 95% CI 3.1–21.7), but was still elevated >10 years after any TBI.

Significance: Herein we present a large population-based case–control study on TBI as a risk factor for unprovoked epileptic seizures, including cases of all ages with individually validated seizure diagnoses. The risk for epileptic seizures was substantially increased after TBI, especially during the first 6 months after the injury and in patients with a combination of ICH and brain contusion.

KEY WORDS: Epidemiology, Seizure, Epilepsy, Traumatic brain injury, Case–control, Population-based.



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Traumatic brain injury (TBI) is a leading cause of epilepsy and estimated to contribute to 4–9% of all epilepsy cases.^{1–3} Given the importance of TBI for the overall incidence of epilepsy, there is a need for population-based studies on risks for developing posttraumatic epilepsy (PTE), which may assist in fine-tuning individual

risk assessment for TBI patients, and also in identifying target populations suitable for future antiepileptogenic interventions.

Most previous reports are cohort studies of selected TBI patients^{4–16} from trauma centers, making it difficult to generalize from these results. A few population-based cohort studies on patients with TBI have been published, but often with a considerable proportion of patients lost to follow-up,^{17,18} or with use of register data without validation of seizure or TBI diagnoses.¹⁹ Of the two major population-based studies, one was mainly based on TBI data obtained before introduction of modern neuroimaging and had 25% dropouts.² The other study²⁰ used the Danish National Hospital Register to identify those with a diagnosis of TBI and a subsequent diagnosis of epilepsy from 1977 through 2002. However, it included only patients younger than 26 years

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KEY POINTS

- This population-based study suggest that the risk of epileptic seizures is increased twofold after mild and ninefold after severe traumatic brain injury (TBI)
- Although the risk after an isolated intracranial hemorrhage or brain contusion is increased five- to sixfold, a combination of these diagnoses leads to a 43 times increased risk
- During the first 6 months after a severe TBI, the risk for epileptic seizures is increased almost 50 times
- The risk for developing epileptic seizures is still increased >10 years after the trauma

and relied on unvalidated registry data for the diagnoses of both epilepsy and TBI.

The Stockholm Incidence Registry on Epilepsy (SIRE) is a population-based registry of 1,885 individually validated cases with incident unprovoked seizures. We used the SIRE and the Swedish National Inpatient Registry (IPR) to carry out a case-control study to examine previous TBI as risk factor for developing unprovoked seizures, quantifying risks in relation to age, gender, severity and type of TBI and time since TBI, thus providing a first large population-based study with individually validated seizure diagnoses, comprising all ages, and from a period after introduction of modern neuroimaging.

METHODS

From September 1, 2000 through August 31, 2008, all patients identified with a first unprovoked seizure or epilepsy in Northern Stockholm, an urban region with 998,500 inhabitants, were included in the SIRE. The SIRE methodology has been described in detail earlier.²¹ In short, potential cases living in the catchment area were identified through several mechanisms. Three regional hospitals, of which only one has departments of neurology, pediatrics, and neurosurgery, serve the inhabitants of the region. All three have outpatient clinics for adult patients with neurologic diagnoses. A reporting network was established, consisting of all neurologists, pediatricians, and geriatricians as well as nurses in nursing homes in the region. Because all electroencephalography (EEG) results from the region were read by one central EEG laboratory, all EEG requests were screened for suspected new-onset seizures. All new referrals to the neurooncology section of the Karolinska University Hospital were screened, as well as all medical records from patients discharged for the first time from the department of Neurology or Pediatrics at the same hospital with the International Classification of Diseases, Tenth Revision (ICD10) diagnoses G40, G41, or R56.8. Pediatric emergency room

records were evaluated to catch cases not reported otherwise. The medical records of all potential cases were evaluated, and potential cases were thus classified based on the available information 6 months after the index seizure by a panel consisting of a neurologist, a neuropsychiatrist, a resident in neurology, a resident in pediatrics, and a trained research nurse. Potential cases were classified as confirmed unprovoked seizure, provoked seizure, nonepileptic seizure (including psychogenic), and cases uncertain whether unprovoked seizure or not. Only those cases considered to have an unprovoked seizure confirmed by the medical chart review were included in the analysis.

Participants

Inclusion criteria was a newly diagnosed unprovoked seizure. All patients with a seizure or epilepsy diagnosis before September 1, 2000 were excluded, except seizure-free patients with 5 years off medication, who were declared in remission. Seizures occurring <7 days after a TBI were defined as acute symptomatic and not included, unless followed by at least one unprovoked seizure >7 days after the TBI.

Of approximately 19,500 EEG requests and 5,000 screened medical records, 1,885 cases were found to fulfill our inclusion criteria. Their medical records were reviewed 6 months after the index seizure by the study group, which classified the cases according to the 1993 and 1997 recommendations of the International League Against Epilepsy (ILAE).^{22,23} Eight controls per case (in total 15,080 controls), matched by sex, catchment area, and year of diagnosis (incidence density sampling²⁴), were randomly selected from the Population Register of Stockholm County.

History of TBI

The IPR provides ICD codes for all inpatient care in Stockholm since 1969. In 1980, hospital discharge diagnoses were reported to the IPR in a majority of Swedish counties, and the registry is considered complete nationwide since registration became mandatory by law in 1987. Cases and controls were linked to this registry using the 10-digit personal identification number assigned to all Swedish citizens, in order to identify a history of in-hospital care for TBI from 1980 to 2008. Diagnostic information was based on the Swedish version of the ICD-8 between 1980 and 1986²⁵ on ICD-9 between 1987 and 1996²⁶ and since 1997 on ICD-10 (WHO 1994).^{27,28}

TBI was defined as a hospital discharge diagnosis of head injury including the following diagnoses: skull fracture (ICD-8, ICD-9: 800, 801, 803; ICD-10: S020, S021, S027, S029), mild TBI (concussion: ICD-8, ICD-9: 850; ICD-10: S060) and severe TBI (ICD-8, ICD-9: 851–854; ICD-10: S061–S069). Severe TBI was further subdivided into brain contusion, including parenchymal hemorrhage (ICD-8, ICD-9: 851; ICD-10: S061–S063), and intracranial, nonparenchymal hemorrhage (ICH, including subdural-,

subarachnoid-, and epidural hemorrhage) (ICD-8, ICD-9: 852; ICD-10: S064–S066).

Comorbidity

Through the IPR, information on diagnoses related to alcohol abuse (ICD-8: 291, 303, 571.00, 571.01, 577.10 ICD-9: 291, 303, 305A, 357F, 425F, 535D, 571A–D, 790D; ICD-10: E244, F10, G312, G621, G721, I426, K292, K70, K852, K860, O354, T51, Y90, Z714, Z721), psychiatric disease (ICD-8: 295–299, 300.1–4, 311; ICD-9: 295–299, 300A–F, 300W, 300X, 311; ICD-10: F20–F34, F38–F42), stroke (ICD-8, ICD-9: 430–438; ICD-10: I60–I69, G45), brain tumor (ICD-8: 191, 225.00–225.20; ICD-9: 225A–C; ICD-10: C71, C793, D33, D43), or dementia (ICD-8: 290.0, 290.1; ICD-9: 331A–F; ICD-10: F00–F02, F039) was collected. The number of patients and controls with these diagnoses that could potentially influence the risk of having a TBI was recorded, and the number of patients with TBI in these subgroups determined.

Statistical analysis

Analyses were conducted using SAS (Version 9.3; SAS Institute Inc., Cary NC, U.S.A.).

We used conditional logistic regression to calculate odds ratios (ORs) and 95% confidence intervals (95% CIs) for a first unprovoked seizure in relation to TBI, adjusting for age.

This case–control study was based on incident cases of epileptic seizures, and the controls were selected by incidence density sampling. In this setting, it has been shown that the odds ratio (OR) provides a valid estimate of the incidence rate ratio and the relative risk (RR),^{29,30} and will therefore be referred to as RR in this article.

RRs and 95% CIs were calculated regarding different severity of TBI. Attributable fraction (AF) was estimated for mild and severe TBI according to the formula $(RR - 1 / RR) * p$, where p is the prevalence of TBI among the cases.²⁴ Analyses were stratified by sex and age (older than 65 years compared to 65 years or younger at TBI years at TBI). RRs and 95% CIs for different combinations of head injuries and within different time intervals after trauma (<0.5 year, 0.5–2 years, 2–10 years, and >10 years) were studied in detail. Patients with more than one hospitalization for head injury were included from the date of the first trauma. RR was estimated in relation to number of hospitalizations for TBI (with a washout period of at least 3 months between the diagnoses to exclude patients with more than one hospitalizations for the same trauma) to examine a possible effect of recurrent TBI on the incidence of late posttraumatic seizures. Median time to onset of unprovoked seizures was estimated after mild and severe TBI. To estimate interaction between combinations of TBI we calculated the Relative Excess Risk due to Interaction (RERI) together with confidence intervals.³¹

Standard protocol approvals, registration, and patient consents

Approval was granted by the ethics review board at Karolinska Institutet, Stockholm, which deemed that no individual informed consent was required.

RESULTS

Characteristics

Demographic data and other characteristics of cases and controls are summarized in Table 1. Number of cases with other diagnoses possibly affecting the RR of TBI and number of those with these comorbidities who actually had a diagnosis of TBI are listed separately.

Relative risk of seizures and seizure relapse in relation to type of TBI

The number of cases/controls and RRs with 95% CIs for developing unprovoked seizures after head injury of different severity are given in Table 2. The RRs were increased for all types of head injuries, and amplified with TBI severity. Although mild TBI was associated with a doubled risk (RR 2.0, 95% CI 1.5–2.7) for unprovoked seizures, the risk increased by the factor 8.1 (95% CI 5.1–12.7) after a severe TBI. The AF due to mild TBI was estimated at 1.8% and for severe TBI at 1.9%.

Stratification by gender showed an increased risk for unprovoked seizures after TBI for both women and men (Table 3). Although the increase was similar for both sexes

Table 1. Characteristics of cases and controls

	Cases	Controls
Total	1,885	15,080
Median age at inclusion in years (IQR)	33 (51)	32 (41)
Female, N (%)	852 (45.2)	6,816 (45.2)
Male N (%)	1,033 (54.8)	8,264 (54.8)
Age 65 years or younger, N (%)	1,507 (79.9)	13,249 (87.9)
Age over 65 years, N (%)	378 (20.1)	1,831 (12.1)
Traumatic brain injury (TBI), N (%)	105 (5.6)	317 (2.1)
Females with TBI, N (%)	39 (4.6)	105 (1.5)
Males with TIB, N (%)	66 (6.4)	212 (2.6)
Median age at TBI in years (IQR)	57 (38)	36 (28)
Median age at severe TBI in years (IQR)	61 (24)	60.5 (49)
Number of patients with comorbidities/number with TBI		
Alcohol abuse	69/16	171/24
Psychiatric comorbidities	81/11	241/15
Stroke	215/27	244/21
Brain tumors	41/1	20/1
Dementia	33/4	36/1

IQR, interquartile range in years.

Age at inclusion refers to age at incident seizure among cases and age at year of matching among controls.

Traumatic brain injury (TBI) including skull fracture (ICD-8, ICD9: 800, 801, 803; ICD-10: S020, S021, S027, S029), mild TBI (ICD-8, ICD-9: 850; ICD-10: S060) and severe TBI (ICD-8, ICD-9: 851–854; ICD-10: S061–S069).

Table 2. Relative risk with 95% confidence intervals for unprovoked epileptic seizures and risk of recurrent seizures by severity of traumatic brain injury

Type of TBI	No. cases	No. controls	RR	95% CI
No history of TBI	1,780	14,763	1 (ref.)	
Skull fracture	18	33	4.5	2.5–8.1
Mild TBI	66	266	2.0	1.5–2.7
Severe TBI	41	38	8.1	5.1–12.7
TBI total ^a	105	317	2.7	2.2–3.4

Cases: cases with unprovoked seizures; RR, relative risk; 95% CI, 95% confidence interval; TBI, traumatic brain injury; ref., reference population.

Skull fracture (ICD-8, ICD-9: 800, 801, 803, ICD-10: S020, S021, S027, S029), mild TBI (ICD-8, ICD-9: 850; ICD-10: S060), severe TBI (ICD-8, ICD-9: 851–854, ICD-10: S061–S069) and TBI total (including skull fracture, mild TBI and severe TBI).

^aLess than the total sum of above-mentioned head injuries, since some of the patients contribute to more than one of the categories.

Table 3. Relative risk with 95% confidence intervals for unprovoked epileptic seizures after the diagnoses mild and severe traumatic brain injury stratified by gender and age at traumatic brain injury

	No. cases	No. controls	RR	95% CI
Mild TBI				
Women	26	84	2.4	1.5–3.8
Men	40	182	1.9	1.3–2.7
≤65 years	49	244	1.8	1.3–2.5
>65 years	17	22	2.5	1.4–4.3
Severe TBI				
Women	14	14	7.58.1	3.5–16.0
Men	27	24	8.4	4.8–14.8
≤65 years	25	21	10.7	6.0–19.1
>65 years	16	17	4.6	2.3–9.2

Cases: cases with unprovoked seizures; RR, relative risk; 95% CI, 95% confidence interval.

Mild TBI (ICD-8, ICD-9: 850; ICD-10: S060); severe TBI (ICD-8, ICD-9: 851–854; ICD-10: S061–S069).

after severe TBI, there was a trend toward a higher increase in women (RR 2.4, 95% CI 1.5–3.8) than in men (RR 1.9, 95% CI 1.3–2.7) after a mild TBI.

A tendency toward a higher increase in RR for unprovoked seizures among patients over 65 years at the time of mild TBI compared with those younger than 65 was observed. In contrast, age >65 years at the time of severe TBI seemed to be associated with a reduced increase in RR for unprovoked seizures compared to age ≤65 years (Table 3).

Relative risk of seizures in relation to type of TBI

Table 4 presents RRs with 95% CIs for various combinations of head injuries. An isolated ICH was associated with an almost five times increased seizure risk and a contusion of the brain with a six times increased risk, whereas a combination of ICH and cerebral contusion increased RR for unprovoked seizures by a factor of 42.6 (95% CI

12.2–148.5). RERI for this combination was estimated at 33.2 (95% CI –20.3 to 86.7). An isolated skull fracture was not associated with an increased risk of epileptic seizures unless accompanied by a severe TBI. After a severe TBI without a skull fracture, seizure risk was increased sixfold, whereas individuals with a combination of severe TBI and skull fracture had a 15 times increased risk and RERI was estimated at 8.4 (95% CI –5.3 to 22.2).

Relative risk of seizures in relation to time since TBI

The RR of developing unprovoked seizures was highest within the first 6 months after severe (48.9, 95% CI 10.9–218.9) and mild TBI (8.1, 95% CI 3.1–21.7), and then declined (Table 5). However, >10 years after TBI, the risk was still elevated after both severe (RR 2.2, 95% CI 0.6–8.0) and mild TBI (RR 2.0, 95% CI 1.4–2.9). The median latency time between severe TBI and onset of unprovoked seizures was 1.6 years, whereas it was considerably longer after mild TBI (9.6 years).

Sensitivity analysis

To check for potential confounding factors that could have influence on both risk for unprovoked seizures and risk of TBI, we excluded persons with prior hospital discharge diagnoses of cerebrovascular lesion (CVL), CNS tumor, dementia, alcohol-related diagnoses, or psychiatric diseases with potential impact on risk of developing epilepsy (depression, psychosis, and anxiety disorders³²) in the IPR. This had only minor influence on the risk of developing unprovoked seizures, for example, the RR after mild and severe TBI was estimated at 1.8 (95% CI 1.3–2.5) and 10.2 (95% CI 5.6–18.7), respectively, after adjustment. Because the influence of these comorbidities on our results were negligible and exclusion of them in our further analyses would have considerable impact on our confidence intervals in our subgroup analyses, we decided to perform further analyses including all patients with TBI.

A few patients had previous hospital discharge diagnoses of repeated mild TBI (8 cases and 17 controls), repeated severe TBI (5 cases and 7 controls), or combinations of mild and severe TBI diagnoses (4 cases and 5 controls). We could not detect any accumulating effect of multiple TBI on the RR for developing unprovoked seizures. Although the RR was 2.6 (95% CI 2.2–3.6) for patients (95 cases and 277 controls) with only one hospital discharge diagnosis of TBI, it was 3.1 (95% CI 1.3–7.3) for patients (7 cases and 19 controls) with two TBI, and after three or more TBI the RR was 2.8 (95% CI 0.3–26.8) (one case and 3 controls). Repeated TBI may be a result of the same disease, and epileptogenesis is likely to start after a single TBI; therefore, we choose to include patients with recurrent TBI diagnoses in our further analyses, using the date of first hospitalization for TBI. Separate analysis after exclusion of these patients, as well as calculating time until index seizure from last trauma instead

Table 4. Relative risk with 95% confidence intervals for unprovoked epileptic seizures following various combinations of skull fracture and severe traumatic brain injury (TBI) respectively after possible combinations of intracranial hemorrhage (ICH) and brain contusion

Type of TBI	No. cases	No. controls	RR	95% CI
Skull fracture only	4	25	1.3	0.5–3.8
Severe TBI only	27	30	6.4	3.7–10.8
Skull fracture and severe TBI	14	8	15.1	6.3–36.4
Brain contusion only	8	11	5.9	2.4–15.0
ICH only	14	20	4.5	2.2–9.0
Brain contusion and ICH ^a	15	3	42.6	12.2–148.5

Cases: cases with unprovoked seizures; RR, relative risk; 95% CI, 95% confidence intervals; TBI, traumatic brain injury.

Skull fracture (ICD-8, ICD-9: 800, 801, 803, ICD-10: S020, S021, S027, S029); severe TBI (ICD-8, ICD-9: 851–854, ICD-10: S061–S069); brain contusion (ICD-8, ICD-9: 851, ICD-10: S061–S063); intracranial hemorrhage (ICH) (ICD-8&9: 852, ICD-10: S064–S066).

^aLess than the previous sum of all severe TBI as the diagnoses (ICD-8&9: 854; ICD-10: S067–S069, four cases and four controls) were not included in this calculation.

Table 5. Relative risk and 95% confidence intervals for unprovoked epileptic seizures by severity and time since traumatic brain injury

Time after injury by severity of TBI	No. cases	No. controls	RR	95% CI
Mild TBI				
0–0.5 years	8	8	8.1	3.1–21.7
0.5–2 years	6	19	2.6	1.0–6.4
2–10 years	19	104	1.5	0.9–2.4
>10 years	33	135	2.0	1.4–2.9
Severe TBI				
0–0.5 years	12	2	48.9	10.9–218.9
0.5–2 years	12	9	10.9	4.6–25.8
2–10 years	14	16	7.1	3.5–14.6
>10 years	3	11	2.2	0.6–8.0

Cases: cases with unprovoked seizures; RR, relative risk; 95% CI, 95% confidence intervals; TBI, traumatic brain injury.

Mild TBI (ICD-8, ICD-9: 850; ICD-10: S060); severe TBI (ICD-8, ICD-9: 851–854; ICD-10: S061–S069).

of first TBI (data not shown), only differed marginally from the presented results.

DISCUSSION

We demonstrate that the risk for unprovoked seizures after mild TBI was doubled compared to the unexposed population, while an isolated ICH or brain contusion led to a five- to sixfold increased risk. The combination of ICH and brain contusion put an individual at 43 times increased risk of later seizures. The risk was highest during the first 6 months following head trauma; after a severe TBI, an

almost 50 times increased risk was observed within this timespan. The risk gradually declined over time, but was still elevated 10 years after both mild and severe TBI.

Strengths and limitations

Strengths of this study include its population-based design and the large number of cases. Seizures were individually and uniformly validated and classified, and information of previous TBI diagnoses was collected through linkage with the IPR and was obtained for a period of up to 28 years. The TBI diagnoses rely on the accuracy of hospital discharge diagnoses and coding.^{33,34} The positive predictive values (PPVs) of IPR diagnoses were estimated to be 85–95% for most diagnoses in earlier studies, with the higher values for more severe diagnoses.³⁵ Although the study of Ludvigsson et al. did not specifically assess the PPV of TBI-specific diagnoses, there is no reason to believe that it should be lower than for the assessed diagnoses or that the rate of misclassification would differ between cases and controls. A substantial misclassification would therefore likely result in an underestimation of the associations between TBI and seizures. Validation of ICD-8 codes for severe TBI diagnoses in Denmark between 1979 and 1996 confirmed diagnoses in 88% of cases.³⁴ We also presume a high accuracy of our severe TBI diagnoses, especially as modern neuroimaging has become more available in later years.

It should also be acknowledged that our categorization of the severity of the TBI is crude and arbitrary, relying entirely on the selected ICD codes, not taking into account clinical markers of severity such as duration of loss of consciousness or injury severity scales. We could neither examine this nor the influence of localization or size of injury on the risk of developing unprovoked seizures, as we did not have access to the medical records of the TBI-afflicted controls. Furthermore, the influence of early posttraumatic seizures on the risk of developing late unprovoked seizures could not be examined, as patients with first seizure within 7 days from TBI were not included in SIRE unless they also had late seizures. Most patients with mild TBI do not attend hospital care for their injuries,³⁶ and the presented data are thus representative only for patients hospitalized for mild TBI. Due to the more complicated clinical course after severe TBI, we assume that practically all patients with these diagnoses were included in our material.

Reverse causation, with an initial unknown epileptic seizure leading to a TBI, could have an influence on the increased risk for unprovoked recurrent seizures seen shortly after the TBI. It is unlikely to explain the increased risk persisting several years after severe TBI though, and has not been mentioned as a problem in earlier studies.²⁰

Because this study was registry based, information on potential confounders including lifestyle and medical history was limited. It is possible that factors such as depression¹⁸ and alcohol abuse,³⁷ which are associated with

increased risk of PTE, also influence the risk of head trauma. In an attempt to elucidate the influence of such factors we adjusted for history of IPR diagnoses of depression, stroke, dementia, alcohol-abuse-related diagnoses, or intracranial tumors. This did not influence our results. Previous analyses of SIRE data have not shown any association between socioeconomic status and occurrence of unprovoked seizures.³²

Interpretation and implications

The association between TBI severity and occurrence of posttraumatic epilepsy is well described and generally accepted.^{34,38} In the population-based Olmsted study,² brain contusion and subdural hematoma were identified as the strongest risk factors for PTE, whereas other factors like skull fractures, older age, and prolonged loss of consciousness were weaker, but significant factors. Other studies have described an association between different radiologic and clinical findings with poorer prognosis according to risk of PTE,³⁹ and there seems to be a relation between amount of focal tissue destruction and development of PTE.⁴⁰

Our results confirm the association between TBI severity and the risk for developing unprovoked seizures. On the other hand, as AF is comparable for mild and severe TBI, both diagnoses roughly contribute equally to the disease burden of epilepsy in society due to the higher prevalence of mild TBI.

The independent effect of ICH and brain contusion on this risk is confirmed, and a tendency toward a synergistic effect is indicated. Old age at TBI (especially age older than 65 years²) was described as a risk factor for PTE,²⁰ and we see a trend towards similar findings after mild TBI, suggesting a higher susceptibility among older patients. After a severe TBI, the RR point estimate was higher for patients who were 65 years or younger at TBI. This is most likely a result of the higher incidence of epilepsy in older patients, which reduces the influence of severe TBI as a relative risk factor for unprovoked seizures. A higher mortality after severe TBI in older patients⁴¹ could also have a similar influence on our results, as well as an under ascertainment of elderly patients by SIRE.²¹

The Danish study by Christensen et al. indicated a higher susceptibility for PTE among women after mild TBI compared to men. Our findings indicated a similar trend, although CIs were overlapping. After a severe TBI, no difference in RR of developing seizures was observed between male and female patients, which is consistent with previous studies.²⁰

The risk for unprovoked epileptic seizures was highest within the first 6 months after mild and severe TBI. Although the median latency between severe TBI and seizure onset was 1.6 years, 30% of all cases with unprovoked seizures after severe TBI had seizure onset within 6 months after the head injury, and 60% within 2 years. The risk eventually declined, but remained increased compared to

the reference population for >10 years after both mild and severe TBI. These findings are in line with earlier studies.^{13,20,42}

Patients with a combination of cerebral contusion and ICH are at very high risk for unprovoked seizures and PTE, which calls for closer monitoring and special precautions in particular during the first years after the trauma. Given their high risk to develop seizures, such patients might also be particularly suitable to target in future studies of potentially antiepileptogenic interventions.

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DISCLOSURE OF CONFLICTS OF INTEREST

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